RATUL HASAN

Robotics, Agentic AI & Intelligent Systems

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Research Interests

Robotics • Agentic AI • CSI • World Models • Computer Vision • Machine Learning • Embodied AI • Multimodal Learning • Edge AI • HRI

Education

BRAC University 2022 – Jul 2026
B.Sc. in Computer Science and Engineering Dhaka, Bangladesh
CGPA: ***/4.00.
Thesis: *ADECS: An Agentic AI Framework for Anomaly Detection and Intelligent Decision Support in Elderly Care System.*
Thesis mark: 98.875/100
Research supervisor: Dr. Md. Golam Rabiul Alam.

Research Experience

Research Assistant 2026 – Present
Department of Computer Science and Engineering, BRAC University Dhaka, Bangladesh

- Conduct research in agentic AI, Wi-Fi CSI sensing, robotics, human–robot interaction, and computer vision under the supervision of Dr. Md. Golam Rabiul Alam.
- Research includes knowledge-guided elderlycare assistance, temporal video anomaly detection, and collaborative work on the reliability of stochastic LLM rankings.

Research Team Lead 2024 – 2025
Biomedical Science and Engineering Research Center (BIOSE), BRAC University

- Led Project 1006: a modular bionic arm integrating Arduino-based control, sensor input, servo actuation, and real-time sensory feedback for prosthetic applications.
- Designed finger-movement and feedback mechanisms and implemented a Python-based 3D hand simulator with Arduino sensor integration and pressure visualization.

Mechanical & 3D Design Member 2022 – 2023
BRACU Duburi, BRAC University Dhaka, Bangladesh

- Designed AUV structural components and payload mechanisms, including torpedo and dropper systems, for autonomous underwater missions.
- Team result: **2nd overall at RoboSub 2023** among 34 international teams; recipient of the **Ingenuity Special Award**.

Research Papers & Manuscripts

[1] **Hasan, R.**, Onon, T. T., Dasgupta, A., Haque, Z. Z., and Alam, M. G. R. (2026). *KAGE: A Knowledge-Guided Edge Expert System for Governed Agentic Assistance in Elderlycare.* In preparation.

[2] Sk, M. S. I., Hossain, K. E., **Hasan, R.**, and Alam, M. G. R. (2026). *Can We Trust Who Wins? Partial Identification of Stochastic LLM Rankings under Unknown Coupling.* 30th International Conference on Artificial Intelligence and Statistics (AISTATS 2027) — submitted.

[3] Onon, T. T., Dasgupta, A., **Hasan, R.**, Haque, Z. Z., and Alam, M. G. R. (2026). *Feature-Cached Temporal Modeling for Multi-Category Behavioral Anomaly Detection in ElderlyCare Video Monitoring.* Engineering Applications of Artificial Intelligence — submitted (2026).

Research Presentations

- *ADECS: An Agentic AI Framework for Anomaly Detection and Intelligent Decision Support in Elderly Care System.* BRAC University Silver Jubilee Conference (25th Anniversary). **Accepted for poster presentation.**
- *Advanced Driver Assistance Systems (ADAS) for Future Mobility.* Poster presentation.

Planned Patent Filing

Knowledge-Guided Edge Expert System with Embodied Dual-Node Environmental Sensing Apparatus for Elderlycare.
Status: Filing planned with DPDT, Bangladesh.

Research & Engineering Projects

ADECS — Agentic Elderlycare System Project
Built an undergraduate research prototype combining video-based behavioral anomaly detection, multi-sensor monitoring, and robotic assistance for intelligent decision support in elderlycare.

Axelia — Autonomous Quadruped Robot Project
Engineered the mechanical structure and embedded motion control for a modular four-legged robot, with walking, turning, sensor feedback, and terrain navigation.

Pixi V1 & V2 — Interactive Humanoid Platform V1 | V2
Built an interactive humanoid platform combining hardware and software for sensor feedback, real-time control, facial recognition, and voice interaction.

Phoenix — Autonomous Mapping Drone Project
Integrated flight control, GPS navigation, and imaging for aerial surveying and mapping, with applications in agricultural and environmental monitoring.

Ovijan V2 & Jol Torongo — Field Robotics Rover | AUV
Developed a rescue rover with environmental sensing, visual monitoring, and a manipulator; built an underwater exploration platform with depth sensing and camera-based observation.

FarmBot — Agricultural Automation
Automated planting, watering, and crop monitoring with Arduino control and smartphone vision, reducing the need for a separate onboard computer.

AURA & Browser Automation — Agentic Software Video
Built a Flutter assistant integrating local LLM inference, memory, and task automation. Developed a Flask/Playwright browser-control service with Gemini-assisted link selection and web information extraction.

Further engineering work: Environmental sensing and cleaning robots, a Theo Jansen walking mechanism, and CO_2 reactor design. **Full portfolio: 113+ projects**.

Competitive Funding

- **UIHP Seed Grant — BDT 90,000.** University Innovation Hub Program support for innovation and prototype development.
- **UIHP Pre-seed Funding — Bionic Arm Team (2025).** Award announced by BRAC University on 11 March 2025 for biomedical innovation.

Honors & Awards

Selected distinctions from **50+ awards and recognitions**.

- **International Gold Award** — IEEE R10-HTC Innovation Challenge (2024), Bionic Arm with Sensory Feedback (team award).
- **Vice Chancellor's Recognition** — BRAC University (2025), recognizing the bdapps Innovation Summit achievement.
- **2nd Runner-Up — bdapps Innovation Summit 2025**, ShooterX; **BDT 50,000 team prize** and platform integration support.
- **National Winner & International Competitor** — IEEE YESIST12.
- **National Champion & Global Finalist** — Technoxian Robotics World Cup.
- **National Winner** — 10 Minute School Science Fair, FarmBot.
- **National 2nd Runner-Up** — IoT & Embedded Systems Hackathon.
- **Regional Winner** — National High School Programming Contest (NHSPC).

Additional recognition: Tech Spectra, RUSC Science Fiesta, 39th National Science and Technology Week, Green Genius, Medico Digital Art Challenge, and Desmos Art Challenge. **Competition participation:** World Robot Olympiad national selection.

Professional Experience & Entrepreneurship

Founder & CEO, Onnesok 2020 – Present
Independent software and technology venture Dhaka, Bangladesh

- Develop and release apps, web platforms, and games serving **500K+ users** across products.
- Lead product development from implementation to release, including TexLab Online, ShooterX, Adventure of Sui, and education-focused applications.
- Delivered hands-on robotics and STEM workshops for children through early Onnesok education initiatives.

Student Developer, BRAC Skill Development Program 2025 – 2026

- Built an AI-assisted career platform for job discovery, candidate matching, and career counseling.

Co-Founder & Former CTO, Roboway Labs 2023 – 2025

- Led robotics, IoT, and software development, including the Pixi humanoid series, CaarMate carpooling app, and hardware–software integration.

Software Platforms & Released Products

TexLab Online — Academic Writing Platform Project
Built a browser-based LaTeX editor with real-time compilation and project management for academic and technical writing, using React and Next.js.

ShooterX & Adventure of Sui — Published Games ShooterX | Sui
Published Flutter games on Google Play, implementing gameplay, progression, leaderboards, and authentication. ShooterX integrates bdapps mobile-balance payments.

Learners, HireMe & Campus Applications LMS | HireMe
Built a learning management system, a service/job platform with real-time chat, and student applications including ROBU App and the BRACU Core course project.

Teaching, Leadership & Service

App Development Instructor 2024 – Present
BRAC University, Residential Semester Bangladesh

- Teach hands-on Flutter and software engineering; mentor students in application architecture and full-stack mobile development.

Lecturer, Learners' Boosters 2025 – 2026

- Delivered practical instruction in app development.

Secretary, Robotics Club of BRAC University (ROBU) 2022 – 2026

- Coordinated robotics projects, technical events, documentation, and club operations.

Facilitator, Ecollab International 2025 – 2026

- Supported international youth climate collaboration and digital communications across Bangladesh, Nigeria, Jordan, Indonesia, and Brazil.

Technical Skills & Training

Programming: Python, C/C++, Dart, JavaScript, TypeScript.
AI & robotics: OpenCV, LLM integration, ROS, Arduino, ESP32, Raspberry Pi, sensor integration, embedded control, CAD, 3D modeling and printing.
Software & systems: Flutter, React, Next.js, Node.js, Flask, Firebase, PostgreSQL, MongoDB, Git, LaTeX, Docker, Linux, Bash, GCP, Azure.
Cloud training: Azure administration, Azure AI Fundamentals, and Azure Fundamentals.

Academic References

Department of Computer Science and Engineering, BRAC University

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